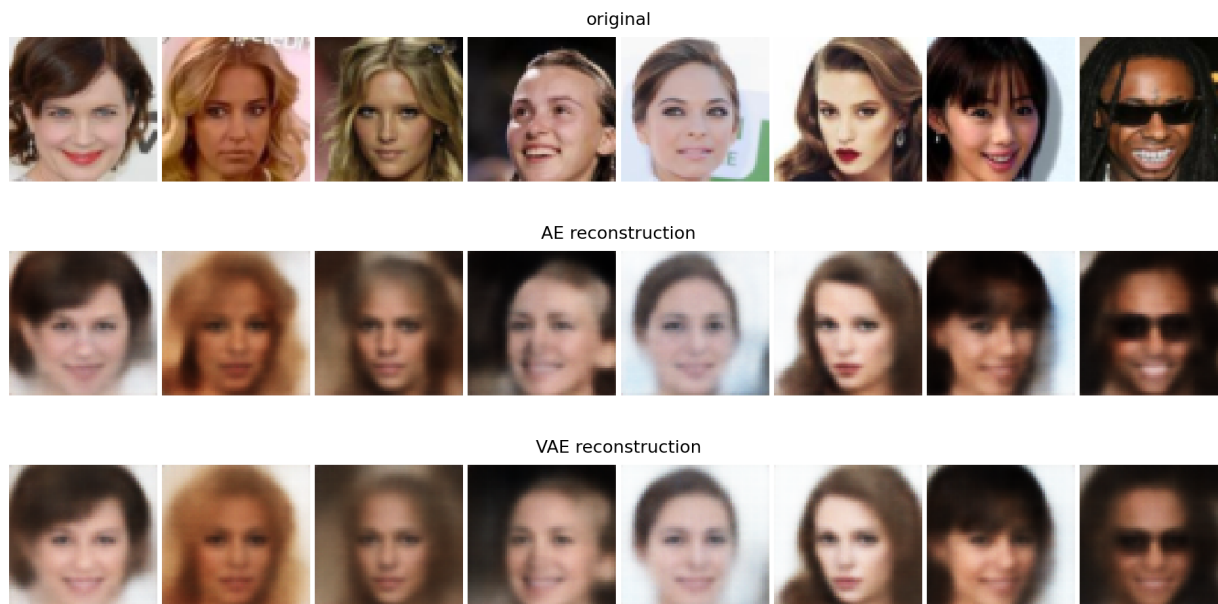


Assignment 1: Generative Models

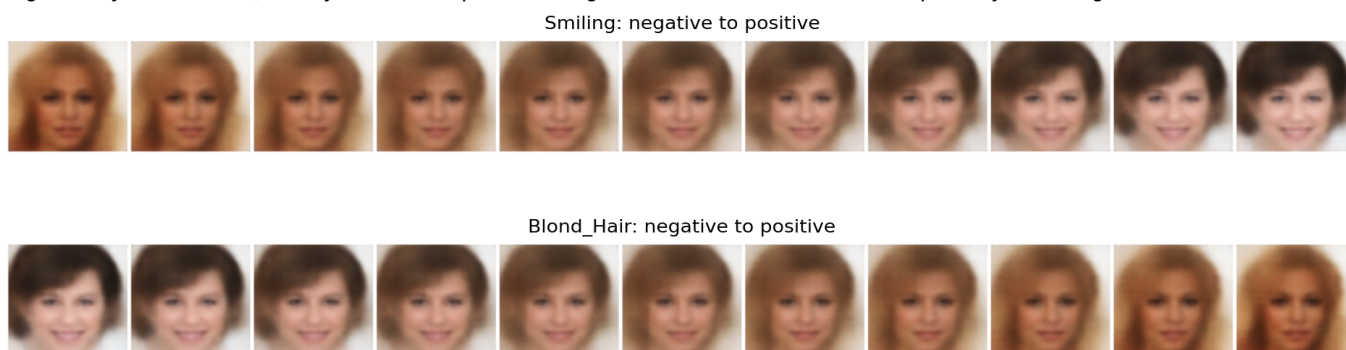
Roll 20L-11327 | held-out digit 7 | digits [0, 2, 3, 6, 7, 9] | CelebA attribute Blond_Hair

A. Autoencoder and variational autoencoder

A1. Both models used the same 27,000/3,000 split, latent size 64, Adam 1e-3, batch 64, and 10 epochs. Final train/validation pixel MSE was **0.006684/0.006605** for the AE and **0.009939/0.008301** for the VAE. The VAE validation MSE is higher by **0.001696**; it is smoother because its latent space is regularised.



A2. Eleven mean-code points remain face-like in both strips. The smile becomes clearer after roughly the middle, while dark hair changes progressively toward blond; identity and hair shape also change, so the attribute direction is not perfectly disentangled.



A3. With the same $N(0, I)$ codes, the AE decoder returns blurred brown averages, whereas the VAE gives varied face-like samples. The AE reconstructs recognisable faces only from its own encoder codes, showing that its unregularised code distribution does not match $N(0, I)$.

B. Denoising autoencoder

Five epochs reduced validation MSE to 0.01272. On one seeded 1,000-image test set, PSNR improved from **13.320 dB** to **19.445 dB**. Denoising is conditional reconstruction toward the learned digit manifold, so it does not require a useful stand-alone sampling prior.

GAN failure experiments

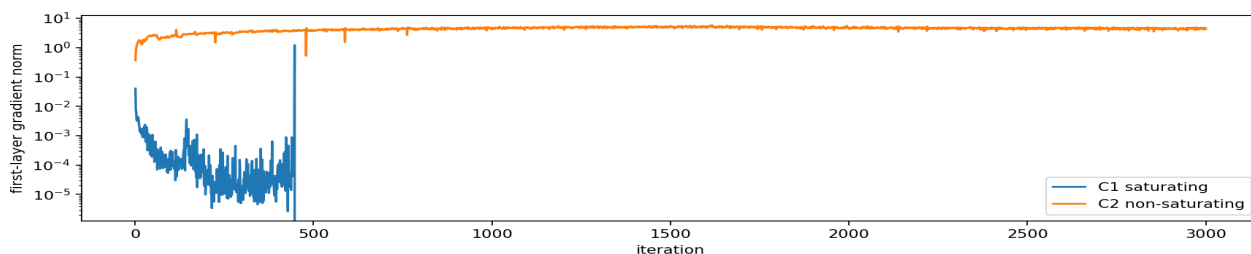
GAN logs contain one row per outer training iteration.

C1. Saturating generator loss

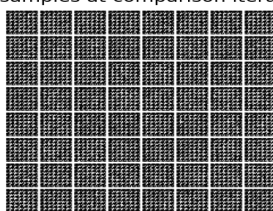
At CSV row/iteration 100 the first-layer generator gradient norm was **7.0218e-05**. The first later value below 1% of that reference was iteration **448**, where the logged first-layer and full-generator gradient norms were both zero. $D(x)$ was 0.790 while $D(G(z))$ was 0.000; the fixed-noise samples remained high-frequency patterns rather than digits.

C2. One-line non-saturating change

Let a be D 's fake-sample logit. The two objectives have the same ideal solution, but $dL_{\text{sat}}/da = -D(G(z))$ while $dL_{\text{non-sat}}/da = -(1-D(G(z)))$. Thus when $D(G(z))$ is near zero, the saturating signal vanishes but the non-saturating signal stays near -1; at iteration 448 its gradient remains strong and the same fixed codes already resemble digits.



C1 samples at comparison iteration



C2 samples at the same iteration



C3. Forced mode collapse

With $z_{\text{dim}} = 2$, no generator BatchNorm, five G steps per D step, and $\text{lr}_G = 2\text{e-}3$, all 5,000 outputs were classified as assigned digit **3**. Modes covered = **1**; reverse KL = **1.79176**. The histogram is exactly [0, 0, 5000, 0, 0, 0].

C3 final



Working and stabilised GANs

Fixed noise is shown at early, one-third, two-thirds, and final checkpoints.

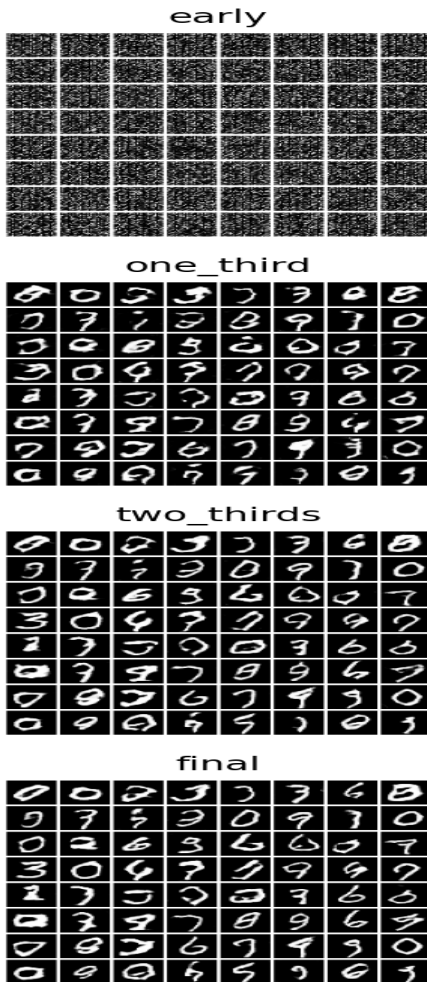
D1. DCGAN

The 1:1 non-saturating DCGAN used BatchNorm in G and after D's second convolution. It moved from noise to recognisable digits and covered all six modes. Reverse KL = **0.05278**.

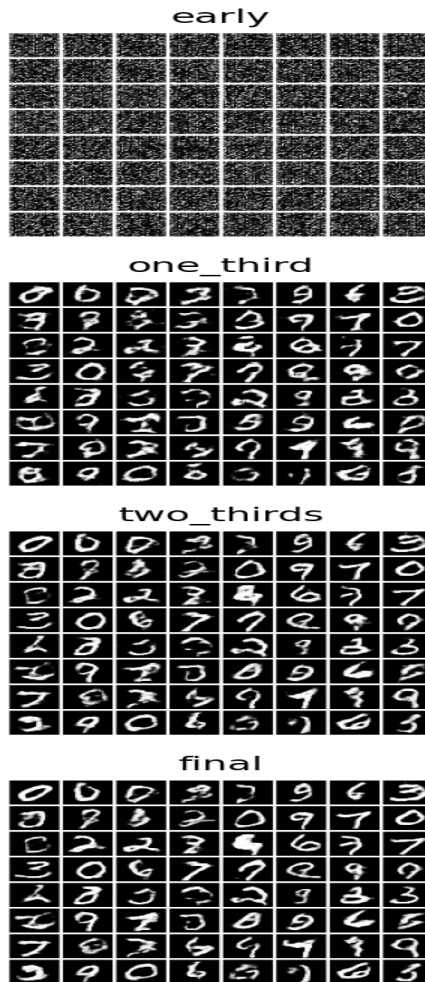
D2. WGAN-GP

The critic used no sigmoid or BatchNorm, five critic steps, and lambda=10 gradient penalty. It also covered all six modes, reduced reverse KL to **0.01980**, and produced cleaner, more balanced digits than D1.

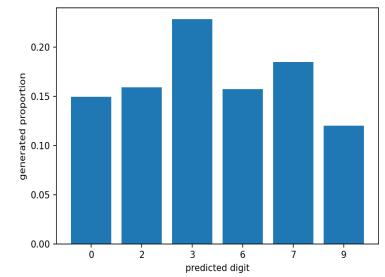
D1 fixed-noise progression



D2 fixed-noise progression



D2 final class proportions



Model	Modes	reverse KL	IS mean	FID
C3	1	1.79176	1.0006	821.241
D1	6	0.05278	4.4772	39.700
D2	6	0.01980	4.6190	17.117

D2 is the strongest generated set on every reported distribution metric: lower reverse KL and FID, slightly higher IS, with all six modes retained.

Evaluation and reflection

Classifier accuracy: 99.019% on the six assigned digits.

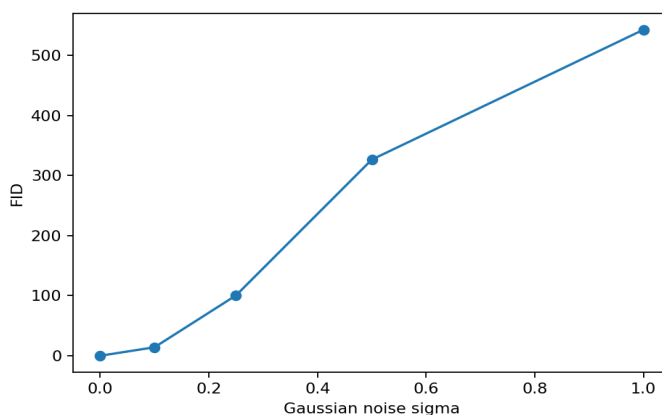
E1-E2. Inception Score and classifier-feature FID

Set	n	IS mean	IS std	FID
real	5000	5.7720	0.0262	1.399
C1	5000	1.0347	0.0011	941.881
C3	5000	1.0006	0.0000	821.241
D1	5000	4.4772	0.1080	39.700
D2	5000	4.6190	0.0898	17.117

C3's IS is 1.0006 because nearly every output receives the same confident class prediction. IS rewards confident predictions and diversity across predicted classes, but does not compare with real data. FID compares generated and real feature distributions, so C3's missing coverage is reflected in its very large FID of 821.241. However, FID reduces those features to a Gaussian's first two moments, so it can miss higher-order structure and artifacts its classifier features do not represent. C1 is farther from real features (941.881), while D2 is much closer (17.117).

E3. Sanity checks

The disjoint real-vs-real FID floor was **1.2009**, measuring finite-sample variation rather than generator error. For sigma [0, 0.1, 0.25, 0.5, 1.0], FID was [0.000, 14.144, 100.552, 326.750, 543.036], a monotonic response to increasing corruption. A nonsymmetric covariance product and floating-point round-off can make sqrtm return a tiny imaginary part; this run's maximum was zero, so taking the real part required no epsilon retry.



Reflection and reproducibility

What surprised me. The saturating gradient reached exactly zero by iteration 448, while the single loss change in C2 produced recognisable digits at that same iteration. WGAN-GP also cut FID by more than half relative to D1 while keeping all modes.

What did not work. C1 never produced digits and C3 intentionally collapsed entirely to digit 3. I first checked saved curves and tensor shapes, then verified ten-image overfits and the GAN smoke test; because those passed, I kept the prescribed failure configurations rather than changing architecture.

LLM use. An AI assistant helped inspect the handout/starter, organise and draft standard PyTorch code, operate the user's Colab run, and assemble this report. Every numerical statement comes from run 20L-11327_main_v1 CSV/JSON evidence; no result was invented.